Portability audit — 2026-10-29

What was already in strong shape

Area	Observation
Tests	10 `tests/` directories (capture, corpus, hypothesis, integration, project, protocol, report, ui, web)
CI	a Python 3.12/3.13 × two-OS matrix, a separate `ui-tests` job offscreen, `code-quality`, `pip-audit`
Packaging	`pyproject.toml` with `gui-scripts`, the `dev` / `web` split, a desktop file, an installer
Documentation	53 documents per language (45 in `docs/`, 8 in the root) + a complete Russian set in `docs/ru/` — a rarity, not the norm
Contracts	`CONTRACT.md`, `PROTOCOL_API.md`, `CAPTURE_API.md`, JSON schemas in `docs/schemas/`
Product invariants	"observation ≠ hypothesis", a mandatory counterexample, reproducibility
Dependencies	the base ones pinned to exact versions (`PySide6==`, `dpkt==`, `PyYAML==`)

Gaps and what was done

#	Gap	Impact	Closed by
1	No `Makefile`	dozens of commands in the README/CI, reproducible only by hand	a `Makefile`: `doctor`, `test-core`, `test-web`, `test-ui`, `wheel`, `package`, `docs-check`, `verify`
2	No "core / interface" split in the check commands	the project could not be checked locally without Qt	`make test-core` (446 tests, ~15 s) against `make test-ui`
3	No host check	a missing `libEGL` surfaced as a PySide6 crash	`scripts/doctor.sh`
4	The build did not prove that the artefact installs	the wheel could be built but not start	`make package` + the `package` job (a clean venv, a headless `--help`, a dependency-set check)
5	53 documents per language without an entry point	a newcomer did not know what to read	`docs/index.md` with a map by question
6	No "claim → check" matrix	"CI is green" was not decoded	`docs/verification/matrix.md`
7	No module specifications	the boundaries were described in the general ARCHITECTURE, not per module	`docs/modules/{capture,protocol}` + a template + a CI check
8	Links pointed at the former repository address (`Pavel1778/...`)	clones and releases did not match the current home	21 links replaced with `ForgeLabHub/ForgeMadrigalProtocolLab`, `make docs-check` catches a regression
9	Nothing checked that the built package does not pull the web/dev dependencies	"I installed it and it brought fastapi"	a check in the `package` job

What remains open

- The `dev`/`web` requirements are given as ranges: strict release reproducibility needs a `pip freeze` snapshot inside the artefact.
- The English and Russian documentation sets could diverge; there is no automatic comparison — closed on 2026-10-30: `tools/docs_pdf.py --check` fails the build when a document has no translation or no PDF.
- The interface tests need PySide6 — outside CI they are available only on a machine with Qt.
- The declared platform is Linux x86-64; Windows/macOS are not supported.